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20 PROMPTS I USE AS A DATA ANALYST

Prompt Engineering Portfolio — Practical AI for Real Analytics Work

EngineeringAbout This Portfolio:

These 20 prompts are the AI-assisted workflows I actively use — or am ready to use — in my daily data analytics work. Each prompt is tool-specific, result-focused, and built to save time, reduce errors, and produce higher-quality output. Covering SQL, Python, Power BI, Tableau, R, Excel, and business communication, they reflect the full analyst stack. This portfolio demonstrates both technical depth and practical prompt engineering skill.

01 EDA Quick-Start Prompt

[Python / ChatGPT / Claude]

WHY THIS PROMPT IS USEFUL: Starting EDA from scratch wastes time. This prompt gets an instant structured analysis plan — missing values, distributions, outliers, correlations — all mapped out before writing a single line of code.

HOW I USE IT IN MY WORK: I paste this into Claude or ChatGPT with my dataset description whenever I get a new CSV or database export. Within seconds I have a 5-step EDA roadmap I directly turn into a Jupyter notebook outline.

THE PROMPT:

"Act as a senior data analyst. I have a dataset with the following columns: [column names & data types]. My business objective is [goal]. Give me a step-by-step EDA plan covering: data quality checks, univariate analysis, bivariate analysis, outlier detection, and correlation insights. Include the Python code snippets I should run at each step."

02 SQL Query Optimiser & Explainer

[SQL / PostgreSQL / ChatGPT]

WHY THIS PROMPT IS USEFUL: Complex joins and subqueries can become slow bottlenecks. This prompt rewrites heavy queries, adds indexes hints, and explains the logic — saving hours of debugging.

HOW I USE IT IN MY WORK: Whenever a PostgreSQL query runs over 10 seconds on large tables, I paste the query here, get an optimised version with EXPLAIN ANALYZE output interpretation, and reduce runtime by 60–80%.

THE PROMPT:

"You are a PostgreSQL expert. Here is my SQL query: [paste query]. The table sizes are: [table_name: X rows]. Optimise this query for performance, explain each change you made, and suggest any indexes I should create. Also explain what the query is doing in plain English for a non-technical stakeholder."

03 Power BI DAX Formula Generator

[Power BI / DAX]

WHY THIS PROMPT IS USEFUL: Writing complex DAX — time intelligence, running totals, cohort metrics — is error-prone. This prompt generates accurate DAX with inline comments so any team member can understand it.

HOW I USE IT IN MY WORK: Used this weekly while building my E-Commerce Sales Dashboard. Generated MTD sales, YoY growth %, and customer retention DAX measures that would have taken half a day to write manually.

THE PROMPT:

"Act as a Power BI expert. I need a DAX measure that calculates [metric description — e.g., Month-over-Month revenue growth %]. My data model has these tables and relationships: [describe model]. Write the DAX formula with inline comments explaining each function, and mention any filter context gotchas I should watch for."

04 Python Data Cleaning Pipeline Builder

[Python / Pandas]

WHY THIS PROMPT IS USEFUL: Dirty data is the 1 bottleneck in analytics. This prompt produces a reusable cleaning script — handling nulls, duplicates, type casting, and outliers — tailored to a specific dataset.

HOW I USE IT IN MY WORK: Applied this during my Customer Shopping Behaviour Analysis project to clean a 50K-row retail dataset. The generated pipeline cut my manual cleaning time from 3 hours to 30 minutes.

THE PROMPT:

"You are a Python data engineer. Here is a sample of my dataset (first 5 rows): [paste sample]. The data issues I've noticed are: [nulls in column X, wrong dtype in Y, duplicate order IDs]. Write a clean, modular Pandas pipeline that handles all these issues, adds a data quality report at the end, and is production-ready with docstrings."

05 KPI Definition & Dashboard Blueprint

[Power BI / Tableau / Excel]

WHY THIS PROMPT IS USEFUL: Stakeholders often say 'build me a dashboard' without defining what matters. This prompt helps me define KPIs, their formulas, owners, and visual best practices before building anything.

HOW I USE IT IN MY WORK: I run this prompt at the start of every dashboard project to align with stakeholders. It gave me a clear KPI tree for my e-commerce dashboard and prevented scope creep.

THE PROMPT:

"Act as a business intelligence consultant. For a [industry/domain] company focused on [business goal], define the top 8–10 KPIs I should track. For each KPI include: definition, calculation formula, recommended visual type in Power BI, target benchmark, and which business question it answers. Output as a structured table."

06 Insight Narrative Generator

[ChatGPT / Claude / Reports]

WHY THIS PROMPT IS USEFUL: Charts without narrative are just pictures. This prompt turns raw numbers into a 3-paragraph executive summary with context, trend, and recommended action — ready to paste into any report.

HOW I USE IT IN MY WORK: After building any dashboard, I extract key numbers and run this prompt to draft the 'Key Findings' section of my reports. Saved 45 minutes per report and improved stakeholder communication.

THE PROMPT:

"You are a data storytelling expert. Here are my analysis findings: [list 5–7 key data points with numbers]. My audience is [C-level / marketing team / ops team]. Write a concise 3-paragraph executive summary that: (1) states the main finding, (2) explains the trend and root cause, (3) recommends a specific action. Keep it under 200 words. No jargon."

07 Statistical Test Selector & Interpreter

[Python / R / Statistics]

WHY THIS PROMPT IS USEFUL: Choosing the wrong statistical test invalidates your analysis. This prompt diagnoses the right test based on data type and sample size, then interprets the output in plain English.

HOW I USE IT IN MY WORK: Used this for hypothesis testing assignments and Deloitte simulation work. Eliminated guesswork around when to use t-test vs ANOVA vs chi-square.

THE PROMPT:

"Act as a statistician. I want to test whether [hypothesis — e.g., 'sales differ significantly by region']. My data: [dependent variable type, independent variable type, sample size,

normality status]. Tell me: (1) which statistical test to use and why, (2) the Python/R code to run it, (3) how to interpret the p-value and effect size in plain English for a business audience."

08 Tableau Chart Type Advisor

[Tableau]

WHY THIS PROMPT IS USEFUL: Picking the wrong chart kills the story. This prompt recommends the best Tableau chart type for a specific data shape and audience, with step-by-step build instructions.

HOW I USE IT IN MY WORK: Whenever I'm unsure whether to use a heat map vs scatter plot vs bar chart for a particular metric, I run this prompt and get a justified recommendation plus Tableau field-drag instructions.

THE PROMPT:

"You are a data visualisation expert specialised in Tableau. I need to visualise [describe your data: X axis has [variable], Y axis should show [metric], coloured by [dimension], audience is [type]]. Recommend the best Tableau chart type with justification, list the exact steps to build it (drag & drop fields), and suggest any formatting best practices to make it executive-ready."

09 Predictive Model Selection Guide

[Python / Scikit-learn]

WHY THIS PROMPT IS USEFUL: Jumping to complex models without justification wastes compute and confuses stakeholders. This prompt selects the right ML algorithm based on problem type, data size, and interpretability needs.

HOW I USE IT IN MY WORK: Used during predictive analytics coursework and personal projects to choose between regression, random forest, and logistic models — with business justifications I could explain to non-technical managers.

THE PROMPT:

"Act as a machine learning consultant. My problem is [classification / regression / clustering]. I have [N rows, M features], the data is [balanced/imbalanced], and my primary requirement is [accuracy / interpretability / speed]. Recommend the top 3 ML algorithms ranked by fit, explain the trade-offs of each, and give a Python Scikit-learn starter code block for the best choice."

10 Excel Formula Translator

[Microsoft Excel / Advanced Excel]

WHY THIS PROMPT IS USEFUL: Complex nested Excel formulas — XLOOKUP, INDEX-MATCH, dynamic arrays — can take 20 minutes to recall and debug. This prompt writes and explains them instantly.

HOW I USE IT IN MY WORK: Weekly use during Excel-heavy reporting tasks. Cuts formula-writing time from minutes to seconds and gives me the explanation to document the formula for colleagues.

THE PROMPT:

"You are an Advanced Excel expert. I need a formula that does: [describe the logic in plain English — e.g., 'look up employee ID in Sheet2 and return their department, but if not found return Unknown']. My data is on [Sheet name], key column is [Column X], return value from [Column Y]. Write the formula, explain how it works cell by cell, and flag any edge cases I should handle."

11 Business Problem to Analytics Brief

[All tools / Planning]

WHY THIS PROMPT IS USEFUL: Analytics without a clear problem statement produces reports nobody reads. This prompt converts a vague business complaint into a structured analytics brief with objectives, metrics, and deliverables.

HOW I USE IT IN MY WORK: I use this at project kickoff — especially for MBA case studies and simulated client briefs — to build a one-page problem statement that aligns data work with business outcomes.

THE PROMPT:

"Act as a senior business analyst. A stakeholder has told me: '[paste their exact vague request]'. Convert this into a structured analytics brief that includes: (1) restated business problem in measurable terms, (2) analytics objective, (3) key questions the analysis must answer, (4) data required, (5) proposed deliverables and timeline. Use a professional tone suitable for a project kickoff deck."

12 Data Model Review & Normalisation Check

[SQL / PostgreSQL / Database Design]

WHY THIS PROMPT IS USEFUL: A poorly normalised schema causes duplicates, wrong aggregations, and maintenance nightmares. This prompt audits a schema and flags violations with fix recommendations.

HOW I USE IT IN MY WORK: Used when designing the database model for project work and reviewing schemas shared by team members. Catches 3NF violations and redundant columns before any query is written.

THE PROMPT:

"You are a database architect. Here is my database schema: [paste CREATE TABLE statements or describe tables/columns]. Review it for: (1) normalisation issues (1NF, 2NF, 3NF), (2) missing foreign keys or indexes, (3) naming convention consistency, (4) any columns that could cause

incorrect aggregations in analytics queries. Provide a prioritised list of fixes with the corrected SQL."

13 Market Analysis Research Synthesiser

[Claude / ChatGPT / Reports]

WHY THIS PROMPT IS USEFUL: Market analysis reports require synthesising data from multiple sources into a coherent narrative. This prompt structures raw market data into a PESTLE or competitive framework instantly.

HOW I USE IT IN MY WORK: Used for MBA market analysis assignments and business case research. Reduced time to produce a structured market brief from 4 hours to 45 minutes.

THE PROMPT:

"Act as a market research analyst. Here is the raw information I have gathered about [industry/market]: [paste notes, stats, bullet points]. Synthesise this into a structured market analysis covering: market size & growth, key trends, competitive landscape, customer segments, and 3 strategic opportunities. Use the PESTLE framework where applicable. Output should be report-ready, 400–500 words."

14 Outlier Investigation Prompt

[Python / SQL / EDA]

WHY THIS PROMPT IS USEFUL: Outliers can mean fraud, data errors, or genuine insights. Blindly removing them destroys value. This prompt creates an investigation framework to decide: remove, cap, flag, or keep.

HOW I USE IT IN MY WORK: Every time I spot suspicious spikes or drops during EDA, I run this prompt to build a 3-step outlier investigation before touching the data — standard practice now in all my projects.

THE PROMPT:

"Act as a data quality analyst. I have found outliers in column [column name] of my dataset. The values are: [list outlier values]. My business domain is [e-commerce / finance / retail]. For each outlier, suggest: (1) probable cause (data entry error / genuine extreme value / fraud signal), (2) recommended treatment (remove / cap / flag / keep), (3) Python code to implement the treatment, and (4) how to document this decision for audit purposes."

15 Automated Report Commentary Writer

[Claude / ChatGPT / Excel / Power BI]

WHY THIS PROMPT IS USEFUL: Monthly reports require the same narrative structure every time. This prompt writes variance commentary — actual vs target, MoM, YoY — in a consistent professional tone at scale.

HOW I USE IT IN MY WORK: I use this at month-end to draft the commentary section of performance reports. Paste in the numbers, get a polished paragraph — then spend time on analysis, not writing.

THE PROMPT:

"You are a financial/business analyst writing monthly performance commentary. Here is the data: Metric: [name], Actual: [X], Target: [Y], Last Month: [Z], Last Year: [W]. Write a 3-sentence variance commentary that: states performance vs target, explains the likely driver of the variance, and recommends a management action. Use confident, concise business language."

16 Python Plot Formatter for Presentations

[Python / Matplotlib / Seaborn]

WHY THIS PROMPT IS USEFUL: Default Matplotlib charts look terrible in presentations. This prompt upgrades any chart with professional fonts, colour palettes, annotations, and export settings ready for PowerPoint or reports.

HOW I USE IT IN MY WORK: I use this to turn raw analysis charts into presentation-grade visuals. Applied on every Python visualisation in my Customer Shopping Behaviour project before adding to the final report.

THE PROMPT:

"You are a data visualisation specialist. Here is my existing Python chart code: [paste code]. Upgrade it to presentation quality by: applying a clean professional colour palette (not default matplotlib), adding a descriptive title and subtitle, annotating the most important data point, removing chart junk (gridlines, spines), setting font to a readable sans-serif, and exporting at 300 DPI. Return the full updated code."

17 Interview Prep: Case Study Solver

[Interview Prep / Business Analytics]

WHY THIS PROMPT IS USEFUL: Data analytics interviews often include case studies with ambiguous data problems. This prompt simulates an interviewer's case question and walks through the STAR + structured analytics answer framework.

HOW I USE IT IN MY WORK: Used intensively while preparing for analytics job applications post-MBA. Helped me structure answers to 'How would you measure churn?' or 'How would you A/B test this feature?' type questions.

THE PROMPT:

"Act as a data analytics interviewer at a top tech/consulting firm. Give me a realistic case study question for a [entry-level / mid-level] data analyst role in [industry]. After I answer, evaluate my response on: problem decomposition, metric selection, data approach, business acumen, and communication clarity. Score each out of 5 and give specific improvement tips."

18 GitHub README Generator for Data Projects

[*GitHub / Portfolio Building*]

WHY THIS PROMPT IS USEFUL: A strong GitHub README is a silent recruiter. This prompt writes a professional, structured README that explains the project purpose, methodology, key findings, and how to run the code — in under 2 minutes.

HOW I USE IT IN MY WORK: Used for both my GitHub projects (E-Commerce Dashboard and Customer Shopping Behaviour Analysis). A good README gets 3x more profile views and demonstrates communication skills alongside technical ones.

THE PROMPT:

"Act as a technical writer for data science projects. Write a professional GitHub README for my project: Project Name: [name]. Tools Used: [list]. What I did: [1–3 sentence description]. Key findings: [2–3 bullet points]. The README should include: Project Overview, Business Problem, Data & Tools, Methodology, Key Insights, How to Run, and Screenshots placeholder. Use clean markdown formatting."

19 R Language Data Analysis Translator

[*R Language*]

WHY THIS PROMPT IS USEFUL: Switching between Python and R mid-project is friction. This prompt translates a Python Pandas analysis workflow into equivalent R (tidyverse/dplyr) code — or vice versa — with explanations.

HOW I USE IT IN MY WORK: Used when academic coursework required R but I had already done EDA in Python. Eliminated the need to relearn R syntax from scratch and kept project momentum.

THE PROMPT:

"You are bilingual in Python and R for data analysis. Here is my Python Pandas code: [paste code]. Translate this line-by-line into equivalent R code using tidyverse/dplyr. For each translated section, add a comment explaining what it does and note any key differences in how Python and R handle this operation (e.g., index handling, NA treatment, data types)."

20 AI-Assisted Stakeholder Presentation Builder

[*Gamma AI / Claude / PowerPoint*]

WHY THIS PROMPT IS USEFUL: Building a data presentation from scratch while under deadline is a productivity killer. This prompt creates a full slide-by-slide outline — with data points, chart recommendations, and speaker notes — in one pass.

HOW I USE IT IN MY WORK: Used with Gamma AI and Claude to build MBA project presentations and Deloitte simulation deliverables. The structured outline meant I only needed to fill in visuals, cutting deck prep time by 60%.

THE PROMPT:

"Act as a data presentation strategist. I need to present the following analysis to [audience type]: [2–3 sentence summary of your findings and data]. Create a 7–10 slide presentation outline that includes: slide title, key message (1 sentence), recommended visual/chart type, 3 bullet points of supporting data, and speaker notes for each slide. The tone should be [executive / analytical / consultative]."
